

policy. This rule will look like: `allow httpd_t tmp_t:file { read }`

To insert this rule into your policy, you will need to compile it and load it.

Modifying the base SELinux policy is not recommended, especially for beginners. Such rules can be compiled in separate policy modules and loaded into the memory. We will come to SELinux modules in a later part of this series.

Access Vector Cache

Just imagine a scenario where you have installed a new application and are unable to execute it. The SELinux default policy in your system prevents access to files and other resources. How do you find out which allow rules are required and which are not? Also, what kind of overhead will the checking of these rules create on system performance?

If you see the `seinfo` command output above, there are more than 80,000 allow rules in the default targeted policy. Checking for multiple subjects while simultaneously accessing multiple objects can create a serious performance bottleneck.

SELinux tackles the performance overhead issue in the traditional manner—by caching rules. An Access Vector Cache is created from rules being looked up into the policy, so that subsequent look-ups can occur from

the AVC (Access Vector Cache). This provides significant performance benefits.

Access Vector Cache (AVC) denial logging gives an idea of why a particular access has been disallowed. Closer examination of these denial logs will enable you to figure out what allow rules need to be inserted into the policy to allow these actions.

Logging is a key feature of SELinux and it is important for security administrators to be able to decipher log messages. In the next series of this article we will explore how SELinux logging occurs and how to use the logs to effectively create allow rules.

Still to come

- SELinux logging
- Policy modules
- Other types of enforcement rules 

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